WTD Analytics

SLOWLY CHANGING DIMENSIONS

SCDs in Databricks

SCD TYPE 1
OVERWRITE

1. INITIAL DATA:

USERID	NAME	CITY	OPERATION	SEQUENCENUM
124	Raul	Oaxaca	INSERT	1
123	Isabel	Monterrey	INSERT	1
125	Mercedes	Tijuana	INSERT	2
126	Lily	Cancun	INSERT	2
123	null	null	DELETE	6
125	Mercedes	Guadalajara	UPDATE	6
125	Mercedes	Mexicali	UPDATE	5
123	Isabel	Chihuahua	UPDATE	5

2. APPLY CHANGES FOR SCD TYPE 1:

```python
dlt.apply_changes(
    target="target",
    source="users",
    keys=["userId"],
    sequence_by=col("sequenceNum"),
    apply_as_deletes=expr("operation = 'DELETE'"),
    apply_as_truncates=expr("operation = 'TRUNCATE'"),
    except_column_list=["operation", "sequenceNum"],
    stored_as_scd_type=1 )
```

3. UPDATED DATA

USERID	NAME	CITY
124	Raul	Oaxaca
125	Mercedes	Guadalajara
126	Lily	Cancun

EXPLANATION

In SCD Type 1, only the latest data is preserved.

SCD TYPE 2
VERSIONING

1. INITIAL DATA:

USERID	NAME	CITY	OPERATION	SEQUENCENUM
124	Raul	Oaxaca	INSERT	1
123	Isabel	Monterrey	INSERT	1
125	Mercedes	Tijuana	INSERT	2
123	Isabel	Chihuahua	UPDATE	5
125	Mercedes	Mexicali	UPDATE	5
125	Mercedes	Guadalajara	UPDATE	6
123	null	null	DELETE	6

2. APPLY CHANGES FOR SCD TYPE 2:

```
dlt.apply_changes(
    target="target",
    source="users",
    keys=["userId"],
    sequence_by=col("sequenceNum"),
    apply_as_deletes=expr("operation = 'DELETE'"),
    except_column_list=["operation", "sequenceNum"],
    stored_as_scd_type=2 )
```

3. UPDATED DATA

USERID	NAME	CITY	__START_AT	__END_AT
123	Isabel	Monterrey	1	5
123	Isabel	Chihuahua	5	6
124	Raul	Oaxaca	1	null
125	Mercedes	Tijuana	2	5
125	Mercedes	Mexicali	5	6
125	Mercedes	Guadalajara	6	null

EXPLANATION

SCD Type 2 maintains a full history, _START_AT and _END_AT indicate its validation period

SCD TYPE 3
ADDING NEW COLUMNS

1. INITIAL DATA:

USERID	NAME	CITY	OPERATION	SEQUENCENUM
124	Raul	Oaxaca	INSERT	1
123	Isabel	Monterrey	INSERT	1
125	Mercedes	Tijuana	INSERT	2
123	Isabel	Chihuahua	UPDATE	5
125	Mercedes	Mexicali	UPDATE	5
125	Mercedes	Guadalajara	UPDATE	6
123	null	null	DELETE	6

2. APPLY CHANGES FOR SCD TYPE 3:

```python
dlt.apply_changes(
    target="target",
    source="users",
    keys=["userId"],
    sequence_by=col("sequenceNum"),
    apply_as_deletes=expr("operation = 'DELETE'"),
    except_column_list=["operation", "sequenceNum"],
    stored_as_scd_type=3,
    track_history=["city"])
```

3. UPDATED DATA

USERID	NAME	CURRENT_CITY	PREVIOUS_CITY	SEQUENCENUM
124	Raul	Oaxaca	null	1
123	Isabel	Chihuahua	Monterrey	5
125	Mercedes	Guadalajara	Mexicali	6

EXPLANATION

SCD Type 3 tracks limited history by storing previous values in additional columns.